

3_popquiz DC circuit, Ohms law

⚠ This is a preview of the published version of the quiz

Started: Mar 22 at 7:03pm

Quiz Instructions



Question 1 14.5 pts

An AC unit has a power of 1,200Watts .

Whats the energy used in Kwh during a day (24hours) about (round to nearest ten)

Suppose the AC unit is on 24 hours

hint: convert W to KW then multiply by the number of hours

☐

30KWh

☐

30,000 KWh

☐

20KWh

☐

12KWh



Question 2 14.5 pts

Building on question 1. In Miami, the KWh is \$0.15 (about).

How much you will pay for your AC unit (for 1 day). About

☐

\$3

☐

\$4.5

☐

\$1.80

☐

\$9



Question 3 14.5 pts

A student makes a simple circuit with 2 AA batteries (1 AA is 1.5V and the student has 2 in series), a red LED, a switch and a resistor. The LED is a semi-conductor and its voltage drop has to be 1.7V . The resistor R is here to protect the LED so the current does not exceed 20mA(0.02A). What is the minimum resistance he needs?

(so find R when $I = 0.02A$). See what I did in class

hint: from top to ground you have: power supply (2 AA) then voltage drop of 1.7V then the voltage drop across the resistor then ground. find voltage drop across resistor. then find I using Ohms law

□

☐

150 ohms

☐

65 ohms

☐

75 ohms

☐

85 ohms

⋮

Question 4 14.5 pts

If I short out a D cell and draw a current of 7 A for 5 s, how many coulombs of charge move through the circuit?

☐

7C

☐

7/5C

☐

5C

☐

35C

⋮

Question 5 14.5 pts

The kilowatt-hour is a unit of

☐

voltage

☐

power

☐

energy



current



Question 6 14.5 pts

A battery operating a large flashlight produces a current of 2 amperes in a light bulb that has a resistance of 4 ohms. The voltage (drop) across the battery is

(hint: the voltage across the battery= voltage across the load since there is only one load)



None of these



2V



0.5V



8V



Question 7 14.5 pts

An electric toaster operating on 120 volts draws a current of 6 amperes. The resistance of the toaster is

Use Ohms law



0.05 ohms



10 ohms



720 oms



20 ohms

Not saved

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